

SMART SENSORS, SMARTER WEB: Capturing And Recording Data Online

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Sensors now play a pivotal role in how environmental changes are perceived, monitored, and addressed. These smart devices capture real-world inputs—temperature, humidity, pressure, moisture—and generate the raw data that powers innovation across

agriculture, healthcare, environmental science, and smart infrastructure. Their value lies not just in data collection but in effective visualisation and application.

Consider a remote agricultural field where soil moisture is tracked continuously, anomalies

are detected in real time, and decisions are made from afar. Integrating sensors with web technologies transforms this into a practical solution. This fusion bridges hardware and software, enabling real-time visualisation, trend analysis, and seamless offline data logging.

Built using a soil moisture sensor and an Arduino board, this system sends data to a web page via the browser's serial API. A dynamic line graph offers real-time updates, with an option to download data in CSV format.

The goal is to highlight how accessible tools—HTML, JavaScript, and Chart.js—can be used to build effective smart sensing systems. Designed for educational, experimental, or applied use, this setup serves as a strong foundation for exploring hardware-software integration.

Fig. 1 shows the tested prototype in which soil sensor is in pot and Arduino is connected to laptop USB for data transfer. Refer to the Bill of Materials table to assemble the system. Fig. 2 shows the web page showing the data in real time with a graph.

Circuit diagram

Fig. 3 shows the physical connection of the system, while Fig. 4 provides the schematic. In both diagrams, the sensor's output signal pin is connected



Fig. 1: Soil sensor in pot and Arduino connected to laptop USB for data transfer

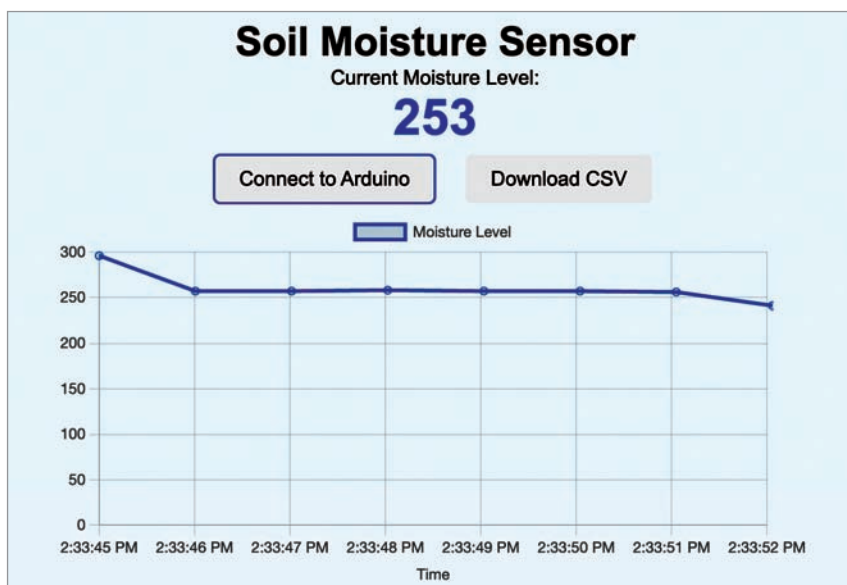


Fig. 2: Web page showing the data in real time with a graph

BILL OF MATERIALS

Components	Quantity
Arduino Uno (MOD1)	1
Soil moisture sensor (S1)	1
USB cable for Arduino	1
Jumper wires (male to female)	As required